



DEPARTMENT OF COMPUTER SCIENCE

COMP438 - Distributed Computing

Course Project - phase 1

Weight: 15 points

Due date: Jun 13, 2026

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Project Summary:

Peer-to-Peer chat application using a Directory server to help clients exchange network credentials (IP and Port numbers).

Project Details:

Your project should be done using Java and it should be organized as follows:

1. Client process:

- Client programs allows user to chat with other clients. Client process establishes TCP connection with Dictionary process (will be explained next) upon activation, it sends a periodic "I am alive" message to the Directory every one second, and Directory saves the IP and Port numbers for that client when receiving "I am alive" message.
- Client process receives a list of all active clients periodically (names if IDs only) from the Directory process, then the user should be able to select a client to chat with, when selected, it asks Directory to send IP and Port numbers for the destination client.
- Client process needs to be multithreaded so it can exchange messages with other clients asynchronously.

2. Dictionary process :

- it links between the clients
- it keeps track of which clients are available.
- it sends the list of available clients periodically (only the names)
- when a client requests a connection with another client, Dictionary sends IP and Port

for destination.

What to Submit?

1. Only .java files for Client, and Dictionary files
2. Report: your report should explain:
 - the design of your project
 - how many sockets needed, how did you manage concurrency
 - explain how client established connection with other clients
 - explain the application layer header your project proposes

Important Instructions:

1. Use Java, do not use any other languages
2. How to submit the project:
 - on ITC ONLY
 - reply only with .java files and the report in pdf
3. The project is a teamwork, each of 2 students
4. There will be a project discussion
5. Any use of AI tools will count as plagiarism
6. Copying codes or code segments from online sources or from any other than your teammate will count as plagiarism

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